

Sea Project — Design System

Single source of truth for the Sea Project's visual language. All values are derived directly from Figma reference screens and the existing project token structure.

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1. Typography

Font family: Nunito

Default letter spacing: -0.25px across all text styles

Word spacing: 0 (normal)

```
$font-family-primary: 'Nunito', system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;
```

1.1 Headline

Style	Size	Line Height	Weight	Letter Spacing
Headline / Large	32px	44px	600 — Semibold	-0.25px
Headline / Medium	28px	38px	600 — Semibold	-0.25px
Headline / Small	24px	33px	600 — Semibold	-0.25px

Usage: Page titles, major section headers, modal titles.

1.2 Title

Style	Size	Line Height	Weight	Letter Spacing
Title / Large	24px	33px	400 — Regular	-0.25px
Title / Medium	20px	27px	500 — Medium	-0.25px
Title / Small	18px	25px	600 — Semibold	-0.25px

Usage: Card titles, section headings, panel headers, navigation labels.

1.3 Label

Style	Size	Line Height	Weight	Letter Spacing
Label / Large	14px	19px	500 — Medium	-0.25px
Label / Medium	12px	16px	500 — Medium	-0.25px
Label / Small	11px	15px	500 — Medium	-0.25px

Usage: Input labels, metadata, tags, badges, tight UI labels, button text (small sizes).

1.4 Body

Style	Size	Line Height	Weight	Letter Spacing
Body / Large	16px	22px	400 — Regular	-0.25px
Body / Medium	14px	19px	400 — Regular	-0.25px
Body / Small	12px	16px	400 — Regular	-0.25px
Body / Small Semibold	12px	16px	600 — Semibold	-0.25px

Usage: Paragraphs, descriptions, form field values, table content, activity feeds.

1.5 Display / KPI

Style	Size	Weight	Usage
Display / Welcome	48px	700 — Bold	Hero greeting, welcome screen headline
Display / KPI 1	64px	700 — Bold	Primary metric values on KPI cards
Display / KPI 2	56px	700 — Bold	Secondary metric values on KPI cards

2. Color System

2.1 Primary — Blue

Token	Value	Usage
--color-primary	#5B8AF0	Default state: buttons, links, active indicators
--color-primary-hover	#4A79DF	Hover state on primary interactive elements
--color-primary-pressed	#3A69CE	Active / while-pressing state
--color-primary-disabled	#AABFF5	Disabled fill (primary buttons)
--color-primary-subtle	#EEF3FD	Light tint background, hover on secondary/tertiary
--color-primary-deep	#4F46E5	Strong emphasis, deep accent

2.2 Semantic

Token	Value	Usage
--color-success	#10B981	Positive status, success banners, green indicators
--color-warning	#F59E0B	Caution alerts, warning badges
--color-danger	#EF4444	Errors, destructive action confirmations
--color-info	#0EA5E9	Informational tooltips, info badges

2.3 Text

Token	Value	Usage
--color-text-primary	#1A2340	Main body text, headings
--color-text-secondary	#5E6E8C	Placeholder, helper text, supporting labels
--color-text-disabled	#A8B8CC	Disabled text on any surface
--color-text-on-primary	#FFFFFF	Text/icons on filled primary backgrounds

2.4 Surface & Background

Token	Value	Usage
--bg-page	#F3F7FB	Page/shell background
--bg-surface	#FFFFFF	Cards, panels, modals, inputs
--bg-subtle	#EEF3FD	Hover tint on secondary/tertiary elements
--bg-base	#0F172A	Dark shell (if dark mode / dark sidebar used)

2.5 Border

Token	Value	Usage
--color-border	#C8D5E8	Default borders, dividers, input outlines
--color-border-focus	#5B8AF0	Focus ring on inputs and interactive elements

3. Spacing

Base unit: 4px . All spacing values are multiples of 4 .

Token	Value	Usage
--space-1	4px	Tight gaps, icon padding
--space-2	8px	Inner component padding, icon-to-label gap

--space-3	12px	Small component padding
--space-4	16px	Standard padding, grid gaps
--space-5	20px	—
--space-6	24px	Section padding, card padding
--space-8	32px	Large section spacing
--space-10	40px	—
--space-12	48px	Page-level vertical rhythm
--space-16	64px	Hero spacing, display sections

4. Border Radius

Token	Value	Usage
--radius-xs	4px	Chips, small tags
--radius-sm	6px	Input fields, small badges
--radius-md	8px	Buttons, dropdowns
--radius-lg	12px	Cards, panels
--radius-xl	16px	Modals, large containers
--radius-2xl	24px	Feature cards
--radius-full	9999px	Pills, avatar rings, toggles

5. Shadows & Elevation

Token	Value	Usage
--shadow-sm	0 1px 2px rgba(26, 35, 64, 0.06)	Subtle lift, table rows
--shadow-md	0 4px 12px rgba(26, 35, 64, 0.08)	Cards, dropdowns, popovers
--shadow-lg	0 8px 24px rgba(26, 35, 64, 0.12)	Modals, drawers, floating panels
--shadow-xl	0 16px 40px rgba(26, 35, 64, 0.16)	Full-page overlays

6. Button System

All buttons support three optional slots: **left icon**, **label text**, and **right icon**.
Border radius on all sizes: `--radius-md` (8px).

6.1 Sizes

Size	Height	Horizontal Padding	Gap (icon ↔ label)	Label Token	Icon Size
XS	28px	10px	4px	Label / Small	14px
S	32px	12px	6px	Label / Medium	14px
M	36px	14px	6px	Label / Large	16px
L	44px	18px	8px	Title / Small	18px

6.2 Primary

Solid filled button. Use for the **single most important action** per view.

State	Background	Text / Icon	Border
Default	#5B8AF0	#FFFFFF	—
Hover	#4A79DF	#FFFFFF	—
While Pressing	#3A69CE	#FFFFFF	—
Disabled	#AABFF5	#FFFFFF	—

6.3 Secondary

Outlined button. Use for **supporting actions** alongside a Primary (e.g. Cancel, Back).

State	Background	Text / Icon	Border
Default	transparent	#5B8AF0	1.5px solid #5B8AF0
Hover	#EEF3FD	#5B8AF0	1.5px solid #5B8AF0
While Pressing	#DCE9FB	#4A79DF	1.5px solid #4A79DF
Disabled	transparent	#AABFF5	1.5px solid #C8D5E8

6.4 Tertiary

Text-only button. Use for **low-emphasis inline actions** (e.g. "View details", "Learn more").

State	Background	Text / Icon	Border
Default	transparent	#5B8AF0	—
Hover	#EEF3FD	#5B8AF0	—
While Pressing	#DCE9FB	#4A79DF	—
Disabled	transparent	#AABFF5	—

6.5 Button Rules

- Never place two Primary buttons side by side.
- Disabled buttons must not receive hover or focus styles.
- Icon-only buttons must include an `aria-label`.
- Maintain consistent padding — do not change horizontal padding per content width.

7. Layout & Breakpoints

Name	Range	Horizontal Padding
Mobile	$\leq 768\text{px}$	16px
Tablet	769px – 1024px	24px
Desktop	$\geq 1025\text{px}$	32px

Grid

- **Mobile:** 4-column grid, 16px gutters
- **Tablet:** 8-column grid, 20px gutters
- **Desktop:** 12-column grid, 24px gutters

8. SCSS Tokens

Paste into `src/styles/variables.scss` :

```
// Typography
$font-family-primary: 'Nunito', system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI',
sans-serif;
$letter-spacing-default: -0.25px;

// Primary
$color-primary: #5B8AF0;
$color-primary-hover: #4A79DF;
$color-primary-pressed: #3A69CE;
$color-primary-disabled: #AABFF5;
$color-primary-subtle: #EEF3FD;
$color-primary-deep: #4F46E5;

// Semantic
$color-success: #10B981;
$color-warning: #F59E0B;
$color-danger: #EF4444;
$color-info: #0EA5E9;

// Text
$color-text-primary: #1A2340;
$color-text-secondary: #5E6E8C;
$color-text-disabled: #A8B8CC;
$color-text-on-primary: #FFFFFF;

// Surface
```

```

$bg-page:    #F3F7FB;
$bg-surface: #FFFFFF;
$bg-subtle:  #EEF3FD;
$bg-base:    #0F172A;

// Border
$color-border:    #C8D5E8;
$color-border-focus: #5B8AF0;

// Radius
$radius-xs:  4px;
$radius-sm:  6px;
$radius-md:  8px;
$radius-lg:  12px;
$radius-xl:  16px;
$radius-2xl: 24px;
$radius-full: 9999px;

// Shadow
$shadow-sm: 0 1px 2px rgba(26, 35, 64, 0.06);
$shadow-md: 0 4px 12px rgba(26, 35, 64, 0.08);
$shadow-lg: 0 8px 24px rgba(26, 35, 64, 0.12);
$shadow-xl: 0 16px 40px rgba(26, 35, 64, 0.16);

// Spacing
$space-1: 4px;
$space-2: 8px;
$space-3: 12px;
$space-4: 16px;
$space-5: 20px;
$space-6: 24px;
$space-8: 32px;
$space-10: 40px;
$space-12: 48px;
$space-16: 64px;

```

9. Layout Patterns

These are the structural building blocks used to compose every page in the Sea Project. Each pattern answers three questions: **what it is**, **when to use it**, and **how it looks in code**.

9.1 App Shell

What: The fixed frame that wraps every authenticated page — topbar, sidebar, and scrollable content area.

When to use: Always. Every page inside the app sits inside the App Shell.

Topbar		56px tall, fixed
Sidebar	Content area	

240px	padding: 24px, background: #F3F7FB
	max-width: 1280px

Region	Value
Topbar height	56px, white bg, 1px bottom border
Sidebar width	240px desktop · 64px collapsed · hidden on mobile
Content padding	24px (desktop), 16px (mobile)
Page background	#F3F7FB
Card surface	#FFFFFF

```
// Quick example – copy this as your base layout
.layout { display: flex; flex-direction: column; height: 100vh; }
.layout__topbar { height: 56px; background: #fff; border-bottom: 1px solid #C8D5E8; }
.layout__body { display: flex; flex: 1; overflow: hidden; }
.layout__sidebar { width: 240px; background: #fff; border-right: 1px solid #C8D5E8; }
.layout__content { flex: 1; overflow-y: auto; padding: 24px; background: #F3F7FB; }
```

9.2 Dashboard / KPI Grid

What: A responsive grid for metric cards at the top of the dashboard, followed by a two-column area for charts and tables.

When to use: Main dashboard, analytics pages, any page that leads with numbers.

Desktop (≥1025px)	[KPI 1] [KPI 2] [KPI 3] [KPI 4] ← 4 columns	
	[Chart – wide] [Feed / Table] ← 2 columns	
Tablet (769–1024px)	[KPI 1] [KPI 2] / [KPI 3] [KPI 4] ← 2 columns	
	[Chart] / [Feed] ← 1 column	
Mobile (≤768px)	[KPI 1] ← 1 column (all stacked)	

```
// KPI row
.kpi-grid {
  display: grid;
  gap: 16px;
  grid-template-columns: repeat(4, 1fr); // desktop
  @media (max-width: 1024px) { grid-template-columns: repeat(2, 1fr); }
  @media (max-width: 768px) { grid-template-columns: 1fr; }
}

// Chart + table row below the KPIs
.content-grid {
  display: grid;
  gap: 16px;
  margin-top: 16px;
}
```

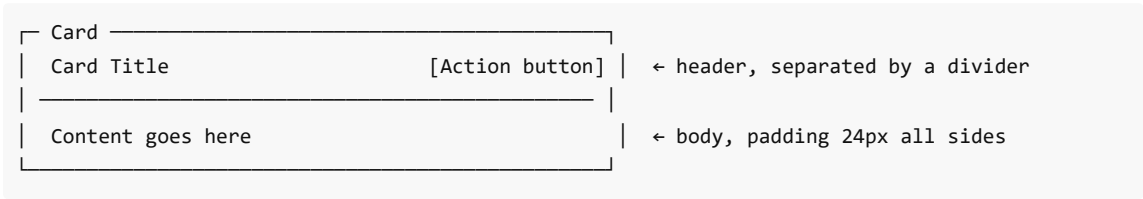


```
grid-template-columns: 1fr 1fr;           // desktop
@media (max-width: 1024px) { grid-template-columns: 1fr; }
}
```

9.3 Card

What: The standard white surface that groups related content. Used for KPI metrics, charts, tables, and activity feeds.

When to use: Any time you need to visually separate a block of content from the page background.



Property	Value
Background	#FFFFFF
Border	1px solid #C8D5E8
Border radius	12px
Shadow	0 4px 12px rgba(26,35,64,0.08)
Padding	24px

```
.card { background: #fff; border: 1px solid #C8D5E8; border-radius: 12px; box-shadow: 0 4px 12px rgba(26,35,64,0.08); padding: 24px; }
.card__header { display: flex; justify-content: space-between; align-items: center; padding-bottom: 16px; margin-bottom: 16px; border-bottom: 1px solid #C8D5E8; }
```

9.4 Two-Column Form Layout

What: A settings-style layout where labels sit in a left column and inputs sit in the right.

When to use: Settings pages, profile pages, multi-field data entry forms.

Section Title

Short description of this group of fields

Full name

[John Smith]

Email address

[john@example.com]

Role

[Plant Manager ▼]

[Cancel] [Save]

- **Left column:** 280px fixed label + helper text
- **Right column:** flex: 1 input controls
- **Row gap:** 24px between each field row

- **On mobile:** stacks to single column, label above input

9.5 Page Header

What: The consistent title block at the top of every content page.

When to use: Every page. Always the first element inside `.layout__content`.

```
← Back to list                (optional, tertiary link)

Plant Overview                (Headline / Large – 32px bold)
Live data for all monitored units (Body / Medium – 14px, grey)
                                [ + Add Unit ] (Primary button)
```

Element	Style
Back link	Tertiary button, size S
Page title	Headline / Large · 32px / 600
Subtitle	Body / Medium · 14px / 400 · #5E6E8C
Action button	Primary, size M, flex-end aligned
Bottom margin	24px before page content

9.6 Stack & Cluster

What: Two micro-layout utilities for arranging items inside a component.

Stack = vertical, evenly spaced items.

Cluster = horizontal, wrapping items (tags, chips, button rows).

```
Stack (vertical)      Cluster (horizontal, wraps)
[Item A]              [Tag 1] [Tag 2] [Tag 3]
[Item B]              [Tag 4] [Tag 5]
[Item C]
```

Variant	Gap	Example use
<code>.stack--tight</code>	8px	Icon + label, stacked form fields
<code>.stack</code> (default)	16px	Card sections, list items
<code>.stack--loose</code>	24px	Page sections
<code>.cluster</code>	8px	Filter chips, tag groups, button rows

```
.stack { display: flex; flex-direction: column; gap: 16px; }
.stack--tight { gap: 8px; }
.stack--loose { gap: 24px; }
.cluster { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
```

10. Error & Edge Case Patterns

Every screen has states beyond the happy path. These patterns define what the UI shows when something goes wrong, is empty, or is loading. Each one follows the same rule: **be clear, be specific, give the user a next step**.

10.1 Inline Form Validation

What: An error message that appears directly below an invalid field.

When: Triggered on field blur or when the user hits Submit.

Example — invalid email field:

Email address *

hello@

← red border: 1.5px solid #EF4444
← background tint: #FFF5F5

⚠ Please enter a valid email address. ← 11px, #EF4444, margin-top 4px

3 rules:

- 1. One message per field — never stack multiple errors.
- 2. Clear the error the moment the value becomes valid (don't wait for re-submit).
- 3. Never rely on color alone — always include the ⚠ icon for accessibility.

10.2 Toast Notifications

What: A small pop-up in the bottom-right corner confirming an action just completed.

When: After save, delete, copy, or any background operation finishing.

4 variants — one per outcome:

Variant	Left border	When to use
✓ Success	#10B981 green	Record saved, file uploaded, action completed
✗ Error	#EF4444 red	Action failed, server error
⚠ Warning	#F59E0B amber	Saved with caveats, approaching a limit
i Info	#0EA5E9 blue	Background task started, non-critical update

Anatomy:

✓ Changes saved
Your profile has been updated

← title: 14px / 600
← body: 12px / 400, grey

width: 320px · shadow-lg · radius-md
position: bottom-right, 24px from edge

Dismiss timing: Success and Info auto-dismiss after 4 seconds . Error and Warning stay until manually closed.

10.3 Page Banner Alert

What: A full-width bar pinned below the topbar for important conditions the user must see before working.

When: Session expiry warning, maintenance notice, missing required setup, permission issue.

Example:

 Your session expires in 5 minutes. Save your work. [\[Renew session\]](#) 

Variant	Background	Text	Border
Error	#FEF2F2	#B91C1C	1px solid #FECACA
Warning	#FFFBEB	#92400E	1px solid #FDE68A
Success	#F0FDF4	#065F46	1px solid #A7F3D0
Info	#EFF6FF	#1E40AF	1px solid #BFDDBE

Height: 48px · single-line · auto for multi-line · border-radius: 6px · placed directly below the topbar.

10.4 Empty State

What: A centered message shown when a list, table, or feed has no data.

When: First-time use (no data created yet), search with no results, or a filter returning nothing.

Template:

[Icon – 64px]

No results found

Try a different search term or clear your filters.

[Clear filters]

← Title/Small · 18px · 600

← Body/Medium · 14px · grey

← Secondary button, size M

Pick the right copy for each scenario:

Scenario	Icon	Title	Button
Search returned nothing		"No results found"	"Clear search" (Secondary)
No records exist yet		"Nothing here yet"	"Create your first record" (Primary)
Filtered to zero		"No matching records"	"Reset filters" (Secondary)
No access		"Access restricted"	"Request access" (Secondary)

10.5 Loading States

What: Visual placeholders while data is being fetched.

Two tools — choose based on what you know:

Tool	Use when	Example
Skeleton	You know the shape of the content (cards, lists, tables)	A grey shimmer rectangle the same size as the card that will load
Spinner	Shape is unknown, or it's a short in-progress action	Button spinner while a form is submitting

Skeleton — how it works:

```
// A shimmering grey placeholder
.skeleton {
  background: linear-gradient(90deg, #E8EDF5 25%, #F3F7FB 50%, #E8EDF5 75%);
  background-size: 800px 100%;
  animation: shimmer 1.5s infinite linear;
  border-radius: 6px;
}
@keyframes shimmer {
  from { background-position: -400px 0; }
  to   { background-position: 400px 0; }
}
```

Spinner sizes: 16px inside a button · 32px for a section · 48px for a full page.

Never show a skeleton and a spinner for the same content area at the same time.

10.6 Network / Server Error

What: A full-page message when a critical API call fails and the page cannot be rendered.

When: 500 errors, network timeouts, failed initial data fetch.

```
[ Error icon - 80px ]

Something went wrong          ← Headline/Small · 24px · 600
We couldn't load this page.   ← Body/Large · 16px · grey
Check your connection and try again.

[ Try again ]                ← Primary button
[Go to Dashboard]            ← Tertiary button
```

Always give two options: **retry** the failed action, and a **safe exit** back to a working page.

10.7 Destructive Action Confirmation

What: A modal that appears before any irreversible action to make sure the user meant to do it.

When: Deleting a record, archiving data, revoking access, clearing all items.

```
| Delete this plant record? | ← Title/Small · 18px
|                           |
```

This will permanently remove all data and cannot be undone.	← Body/Medium · 14px · grey
[Cancel] [Delete]	← Secondary · Danger (red)

width: 400px · radius: 16px · overlay: rgba(26,35,64,0.48)

3 rules:

1. **Cancel is always on the left.** The safe option comes first.
2. **The confirm button is red** (#EF4444), never the primary blue.
3. **Escape key = Cancel.** Clicking outside the modal also cancels.

10.8 Disabled vs Read-Only

What: Two distinct states for controls that cannot be edited — they mean different things and look slightly different.

	Disabled	Read-only
Meaning	Cannot interact with this right now	Data is shown for reference only
Background	#F3F7FB	#F3F7FB
Text color	#A8B8CC (faded)	#1A2340 (full strength)
Border	#C8D5E8 (default)	#C8D5E8 (default)
Cursor	not-allowed	default
Example	A "Save" button before the form is valid	A field showing a system-generated ID

Rule: If a control is disabled for a non-obvious reason, always show a tooltip explaining why.